Part A · Superposition (v2 demo)

Ghosh & A K Chakraborty, *Circuit Theory and Networks* 4e, ch. 5, Solved Problems 5.1–5.14 · new How / What / Why format

HOW TO READ THIS VERSION

- **How:** the literal working, so you can follow the solution line by line.
- **What:** one line saying what we did and where: "KCL at node X", "switch off the 2 A".
- **Why:** the reasoning behind the move. This is the part that carries over to questions you haven't seen.

Fading: A1–A3 label every step. From A4 on, only the steps that use an idea from the **What's new** box get What / Why; routine working is left bare. Practice questions are bare: by then the moves are yours.

KVL, KCL, Ohm's law (you have these) → Nodal / mesh: all sources at once →
Superposition: one source at a time → Thevenin / Norton (next)

Where this sits. You can already solve any circuit with nodal or mesh analysis, taking every source at once. Superposition breaks the same job into smaller circuits, **one source at a time**, which is often quicker by hand. It is also the reason Thevenin's theorem works, which comes next.

Three terms first.

- **Linear circuit:** made of R, L, C, linear dependent sources and independent sources. Doubling every source doubles every current and voltage.
- **Independent source:** a fixed voltage or current (a battery, a 2 A source, 10 cos 4t).
- **Dependent source** (diamond symbol): its value follows some other voltage or current in the circuit, for example $2V_x$.

Statement. In a linear circuit with several independent sources, any current or voltage is the **algebraic sum** of the values caused by **each independent source acting alone**.

Switching a source off means setting its value to zero:

Source	Switched off it becomes	Why
Ideal voltage source	**short circuit**	0 V across it
Ideal current source	**open circuit**	0 A through it
Dependent source	**never switched off**	it is not a source on its own; it follows the circuit

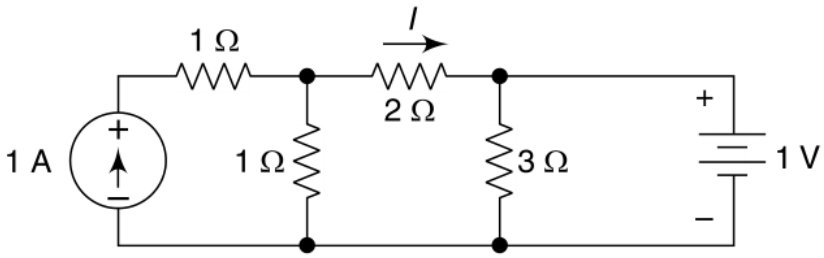
Superposition adds currents and voltages only, **not power** (P goes as I^2, and $(I' + I'')^2 \neq I'^2 + I''^2$).

A1 · Problem 5.1

Superposition · DC **teach · every step labelled**

QUESTION

Find the current I in the circuit shown, using the superposition theorem.



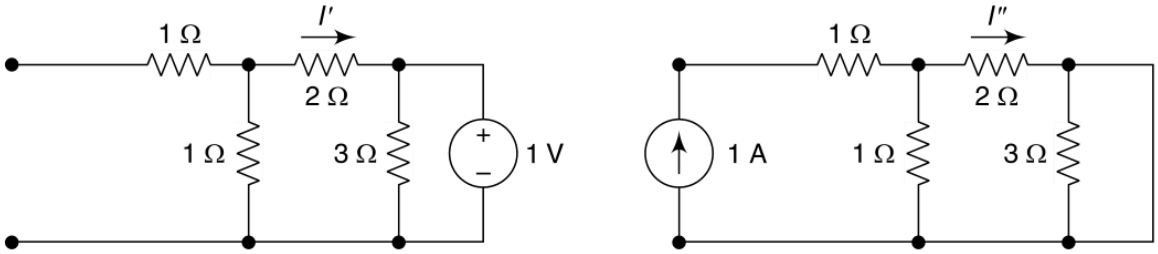
I flows left to right through the 2 Ω resistor.

TRY IT FIRST

Cover everything below. Draw two circuits, 1 V alone and 1 A alone, find the current in the 2 Ω in each, and decide each one's sign before adding.

HINT: WHAT'S NEW

- **The whole method:** two sources, so solve **two small circuits** and add.
- **Switching off:** the 1 V source becomes a short; the 1 A source becomes an open.
- **Signs:** measure each part against the arrow drawn for I. A part flowing the other way is negative.



(i) 1 V acting alone (1 A opened) · (ii) 1 A acting alone (1 V shorted)

Given: 1 A source, 1 V source; 1 Ω (series), 1 Ω (shunt), 2 Ω, 3 Ω. **To find:** I through the 2 Ω (left → right).

HOW

I = I′ (1 V alone) + I″ (1 A alone),
both measured left → right through the 2 Ω

WHAT Split the circuit into two single-source circuits; agree one reference arrow for both parts.

WHY Two sources at once would need simultaneous equations. One source at a time, each circuit is plain series/parallel. The parts only add correctly if both are measured along the **same** arrow, so fix that arrow (I's) before starting.

Case (i): 1 V alone

HOW Replace the 1 A source with an open circuit. The left 1 Ω now ends at an open gap, so it carries 0 A.

WHAT Switch off the 1 A (current source → open).

WHY "Off" means value = 0. A source forcing 0 A behaves exactly like a broken wire. Anything **in series** with that gap carries nothing, so the left 1 Ω drops out.

HOW The 3 Ω is directly across the 1 V, so ignore it. What is left is one loop, 1 V → 2 Ω → 1 Ω (shunt) → back:

$$\text{current} = \frac{1}{2+1} = \frac{1}{3}\ \text{A}, \qquad \text{flowing right} \rightarrow \text{left in the 2 } \Omega$$

WHAT Drop the 3 Ω, then Ohm's law on the single series loop.

WHY An element **in parallel with an ideal voltage source** cannot change anything else: the source fixes its voltage, and it fixes it whether the 3 Ω is there or not. Removing it leaves one series loop, where I = V ÷ (total R).

HOW

$$I' = -\frac{1}{3}\ \text{A}$$

WHAT Give the part its sign by comparing with I's arrow.

WHY The 1 V pushes current out of its + terminal (top right) leftwards through the 2 Ω, opposite to I's arrow. A part flowing against the reference arrow enters the sum as negative.

Case (ii): 1 A alone

HOW Replace the 1 V with a short. The 3 Ω now has a wire across it, so it has 0 V and carries 0 A.

WHAT Switch off the 1 V (voltage source → short); the 3 Ω is shorted out.

WHY A source forcing 0 V is a plain wire. Current always takes a 0 Ω path over a resistor beside it, so a resistor with a wire across it drops out.

HOW

$$I'' = 1 \times \frac{1}{1+2} = \frac{1}{3}\ \text{A} \qquad (\text{left} \rightarrow \text{right, along I})$$

WHAT Current divider at the middle node, between the 1 Ω shunt and the 2 Ω.

WHY The 1 A arrives at the middle node and has two ways back down: the 1 Ω, or the 2 Ω then the short. Two parallel paths sharing a **known** current call for a divider: share in one branch = I × (other branch) ÷ (sum). The series 1 Ω on the left doesn't matter: the source delivers 1 A through it whatever its value.

HOW

$$I = I' + I'' = -\frac{1}{3} + \frac{1}{3} = 0$$

WHAT Add the signed parts.

WHY This is the superposition statement: the full response is the algebraic sum of the single-source responses.

ANSWER

I = 0 A (= book)

WHY ZERO, AND TWO SHORTCUTS

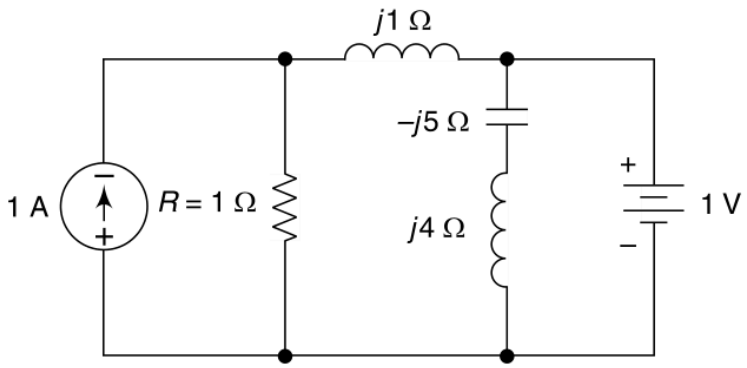
- Node check: $1 = \frac{V_A}{1} + \frac{V_A - 1}{2}$ gives $V_A = 1$ V, the same as the right-hand node, so no current flows in the 2 Ω.
- An element **in series with an ideal current source** never changes any other current (the left 1 Ω here).
- An element **in parallel with an ideal voltage source** never changes any other current (the 3 Ω here).

A2 · Problem 5.2

Superposition · AC **teach · every step labelled**

QUESTION

Calculate the voltage V across the resistor R by using the superposition theorem.



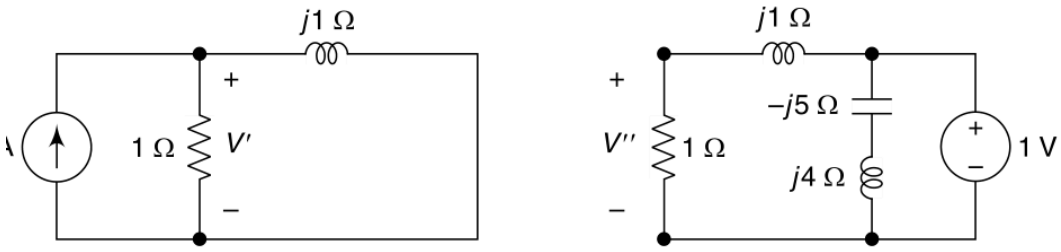
The book draws +/− signs inside the current source. Read it by its arrow: 1 A pushed up into the top node.

TRY IT FIRST

Which elements drop out in each case? Aim for V′ and V″ in the form $\frac{\text{something}}{1 + j}$.

HINT: WHAT'S NEW

- **AC (phasor) circuit.** j1 Ω is an inductor, −j5 Ω a capacitor. Superposition is unchanged; the arithmetic uses complex numbers.
- **Complex dividers:** the same formulas as for resistors, with Z in place of R.
- The −j5 + j4 branch is **across the 1 V source** (5.1's shortcut), so it never matters.



(i) 1 A acting alone · (ii) 1 V acting alone

Given: 1 A, 1 V; R = 1 Ω, j1 Ω, −j5 Ω, j4 Ω. **To find:** V across R (+ at top).

HOW V = V′ (1 A alone) + V″ (1 V alone), both + at the top of R

WHAT Two single-source circuits; one reference polarity (+ at top of R).

WHY Same plan as 5.1. Being AC changes only the arithmetic: every impedance is a complex number, and every rule (Ohm, KCL, dividers, superposition) works on it unchanged.

Case (i): 1 A alone

HOW Short the 1 V. The right top node is now joined to the bottom, so the −j5 + j4 branch has both ends on the bottom wire: it carries nothing. The j1 now runs from the top node straight to the bottom.

WHAT Switch off the 1 V, then redraw to see what is left: R and j1, both from the top node to the bottom.

WHY A branch whose two ends land on the same node has 0 V across it and drops out. After any switch-off, look for elements that now sit between the same two nodes; that is how a messy circuit turns into a simple one.

HOW

$$V' = 1 \times (R \parallel j1) = 1 \times \frac{1 \times j1}{1 + j1} = \frac{j}{1 + j}\ \text{V}$$

WHAT Ohm's law on the parallel pair: $V = I \times Z_{parallel}$.

WHY The 1 A can only go through R or j1, and both join the same two nodes, so they are in parallel. A known current into a known impedance gives the voltage directly; no divider is needed, because we want the voltage, not a branch current.

Case (ii): 1 V alone

HOW Open the 1 A. Drop the −j5 + j4 branch (it sits across the 1 V). One loop remains: 1 V → j1 → R → back.

$$I'' = \frac{1}{1+j} \quad \Rightarrow \quad V'' = 1 \times I'' = \frac{1}{1+j}\ \text{V}$$

WHAT Switch off the 1 A, drop the branch across the source, then Ohm's law on the series loop and on R.

WHY These are 5.1's two shortcuts again (series with a current source, parallel with a voltage source). I″ flows down through R, top to bottom, so the top of R is +, which matches the reference: no sign flip.

HOW

$$V = \frac{j}{1+j} + \frac{1}{1+j} = \frac{1+j}{1+j} = 1\ \text{V}$$

WHAT Add over the common denominator.

WHY Leaving each part as a fraction over the same (1 + j) makes the sum one line. Converting to decimals first would hide the clean answer.

ANSWER

V = 1∠0° V (= book)

CHECK

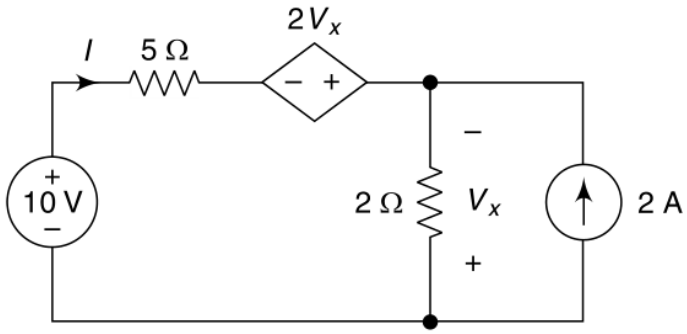
Node equation at the top node: $1 = \frac{V}{1} + \frac{V-1}{j1}$ gives V(1 + j) = 1 + j, so V = 1 V ✓.

A3 · Problem 5.3

Superposition · dependent source **teach · every step labelled**

QUESTION

Use the superposition theorem on the circuit shown to find I.



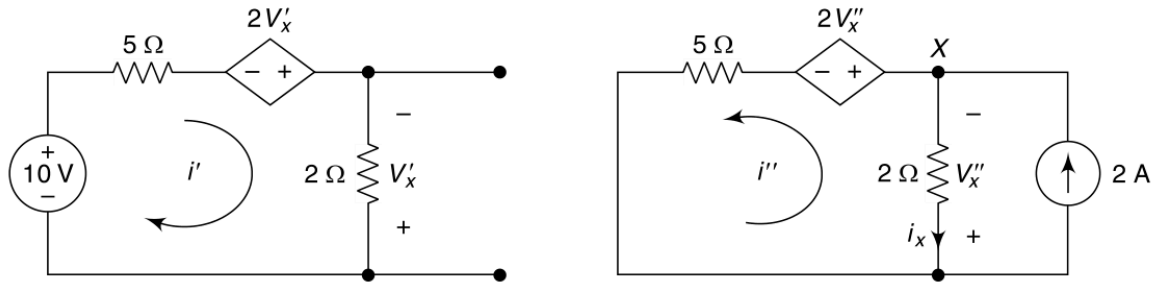
$2V_x$ is a voltage-controlled voltage source. V_x is across the 2 Ω with + at the bottom.

TRY IT FIRST

Write V_x' in terms of the loop current in case (i). For case (ii), write one KCL at the top of the 2 Ω and one equation for the left branch. Expect elevenths.

HINT: WHAT'S NEW

- **Dependent source:** it **stays in every case**. Only the 10 V and the 2 A take turns.
- **Its value changes per case:** $2V_x'$ in case (i), $2V_x''$ in case (ii). Re-find V_x in each sub-circuit.
- **Polarity of V_x:** + is at the **bottom** of the 2 Ω. Most wrong answers to this question come from this sign.



(i) 10 V alone (2 A opened) · (ii) 2 A alone (10 V shorted); the dependent source stays in both

Given: 10 V, 2 A, 5 Ω, 2 Ω, VCVS $2V_x$. **To find:** I (out of the 10 V source, through the 5 Ω).

HOW

$$I = I' \text{ (10 V alone)} + I'' \text{ (2 A alone)}$$

The diamond stays in both circuits, as $2V_x'$ in case (i) and $2V_x''$ in case (ii).

WHAT Two single-source circuits; the dependent source is kept in both.

WHY Superposition switches off **independent** sources only. The diamond has no value of its own; it is "2 × whatever V_x is right now". Each case has its own V_x, so the diamond is the same element with a different value. Shorting it would delete a part of the circuit, not a source.

Case (i): 10 V alone

HOW Open the 2 A. One loop is left: 10 V → 5 Ω → diamond → 2 Ω → back. Call its current i′, clockwise.

WHAT Switch off the 2 A; name one loop current, clockwise.

WHY One loop needs one unknown current. Clockwise matches I's arrow (out of the 10 V's +, rightwards through the 5 Ω), so $I' = i'$ with no sign to fix at the end.

HOW

$$V_x' = -2i'$$

WHAT Write the controlling voltage V_x' in terms of the loop current.

WHY The diamond brings a second unknown, V_x'. **Every dependent source needs this one extra line:** its controlling quantity in terms of your unknowns. i' flows down the 2 Ω, which makes the top of the 2 Ω +. V_x is defined with + at the bottom, so $V_x' = -(2i')$.

HOW

$$10 + 2V_x' = 5i' + 2i'$$

WHAT KVL around the loop, going clockwise.

WHY One loop gives one equation. Going clockwise you **rise** 10 V (− to + through the source), **drop** $5i'$, **rise** $2V_x'$ (the diamond is crossed − to +), and **drop** $2i'$. Total rise = total drop.

HOW

$$10 + 2(-2i') = 7i' \quad \Rightarrow \quad 10 = 11i' \quad \Rightarrow \quad I' = i' = \frac{10}{11}\ \text{A}$$

WHAT Substitute $V_x' = -2i'$ into the KVL and solve.

WHY Now there is one equation in one unknown.

Case (ii): 2 A alone

HOW Short the 10 V, so the left end of the 5 Ω is on the bottom wire. Take the bottom wire as 0 V. Then the top of the 2 Ω (node X) is at $V_X = -V_x''$.

WHAT Switch off the 10 V; choose the bottom wire as reference (0 V).

WHY With a reference node, every point has a single voltage and each branch current is just (voltage difference) ÷ R. X is the − end of V_x and the + end is at 0 V, so X sits at $-V_x''$.

HOW Unknowns: V_x'', and i'' = the current leaving X to the left (through the diamond and the 5 Ω). The current down the 2 Ω is $\frac{V_X}{2} = -\frac{V_x''}{2}$.

WHAT Label the two ways the 2 A can leave node X.

WHY The 2 A arrives at X and has exactly two exits: down the 2 Ω, or left through the diamond and 5 Ω. The diamond needs V_x'' anyway, and the 2 Ω current is already written in V_x'', so two unknowns cover the whole circuit.

HOW

$$\text{KCL at X:} \qquad 2 = -\frac{V_x''}{2} + i'' \qquad \ldots(1)$$

WHAT KCL at node X (top of the 2 Ω).

WHY X is the one node where the **known** 2 A meets **both** unknowns, so in = out gives a single equation containing exactly our two unknowns. General habit: write KCL at the node where the known current source enters.

HOW Walk the left branch from X to ground. X is at $-V_x''$. Crossing the diamond from + to − drops another $2V_x''$, so the right end of the 5 Ω is at $-3V_x''$; its left end is at 0 V.

$$i'' = \frac{-3V_x'' - 0}{5} = -\frac{3}{5}V_x'' \qquad \ldots(2)$$

WHAT Follow the voltage along the left branch (KVL), then Ohm's law on the 5 Ω.

WHY Two unknowns need a second equation. The left branch is a single path, so its current is set by Ohm's law on the 5 Ω once you know the voltage at both ends. Walking from a known voltage and adding each element's rise or drop gives that end voltage.

HOW

$$\text{(2) in (1):} \qquad 2 = -\frac{V_x''}{2} - \frac{3V_x''}{5} = -\frac{11}{10}V_x'' \qquad \Rightarrow \qquad V_x'' = -\frac{20}{11}\ \text{V},$$

$$i'' = \frac{12}{11}\ \text{A}$$

WHAT Substitute and solve.

WHY (2) gives i″ in terms of V_x'', so (1) becomes one equation in one unknown.

HOW

$$I'' = -\frac{12}{11}\ \text{A}$$

WHAT Sign against I's arrow.

WHY i″ flows **leftwards** through the 5 Ω; I's arrow points right. Opposite direction, so the part is negative.

HOW

$$I = I' + I'' = \frac{10}{11} - \frac{12}{11} = -\frac{2}{11}\ \text{A}$$

WHAT Add the signed parts.

WHY Superposition: the sum of the single-source responses, all measured along I's arrow.

ANSWER

I = −2/11 A ≈ −0.182 A (= book). The minus sign means the current actually flows into the + of the 10 V source.

NEVER SWITCH OFF A DEPENDENT SOURCE

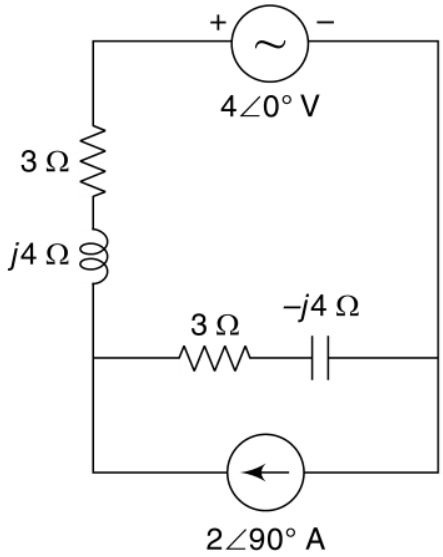
Shorting $2V_x$ as if it were independent gives a wrong answer. It stays in; only its value changes from case to case.

A4 · Problem 5.4

Superposition · AC **new idea · labels on new steps only**

QUESTION

Determine the current in the capacitor branch by the superposition theorem.



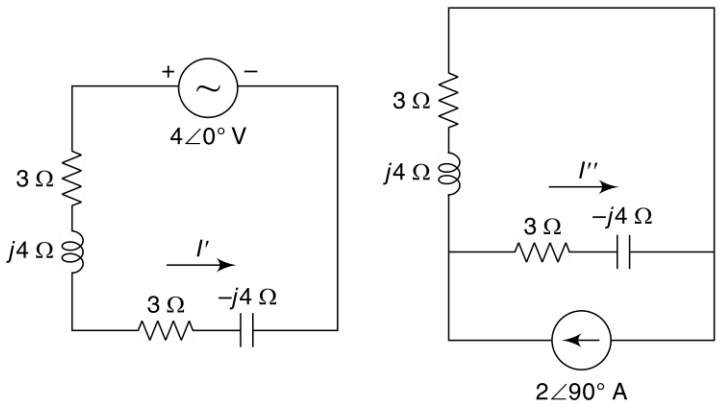
$Z_1 = 3 + j4\ \Omega$ (left branch); $Z_2 = 3 - j4\ \Omega$ (capacitor branch); current source $2\angle 90°$ A.

TRY IT FIRST

Which way does each source see Z_1 and Z_2: in series, or in parallel?

HINT: WHAT'S NEW

- **Source with an angle:** $2\angle 90°$ A = j2 A. Convert to rectangular form before multiplying.
- **Polar form with a negative real part:** a calculator's $\tan^{-1}$ lands in the wrong quadrant; add 180°.



(i) 4∠0° V alone · (ii) 2∠90° A alone

Given: 4∠0° V, 2∠90° A, $Z_1 = 3 + j4\ \Omega$, $Z_2 = 3 - j4\ \Omega$. **To find:** I in the capacitor branch (left → right).

Case (i): 4∠0° V alone (current source opened). Z_1, Z_2 in series:

$$I' = \frac{4\angle 0°}{(3 + j4) + (3 - j4)} = \frac{2}{3}\angle 0°\ \text{A}$$

Case (ii): 2∠90° A alone (voltage source shorted). $Z_1 \parallel Z_2$ across the source:

HOW

$$2\angle 90° = j2, \qquad I'' = j2 \times \frac{3 + j4}{6} = \frac{j6 - 8}{6} = -\frac{4}{3} + j1\ \text{A}$$

WHAT Convert the source to rectangular form, then use the current divider.

WHY Multiplying j2 × (3 + j4) is one line in rectangular form; mixing polar and rectangular in one product is where slips happen. (A number at ∠90° is purely imaginary: 2∠90° = 0 + j2.)

$$I = \frac{2}{3} - \frac{4}{3} + j1 = -\frac{2}{3} + j1\ \text{A}$$

HOW

$$|I| = \sqrt{((2/3)^2 + 1)} = 1.20\ \text{A}, \qquad \theta = 180° - \tan^{-1}(1.5) = 123.7°$$

WHAT Convert to polar, fixing the quadrant.

WHY Real part negative, imaginary part positive, so the point is in the **2nd quadrant** (90° to 180°). A calculator's $\tan^{-1}(1 \div (-2/3))$ returns −56.3°, a 4th-quadrant angle, because tan alone can't tell (−, +) from (+, −). Always read the signs first, then pick the angle.

ANSWER

I = 1.2∠123.7° A (= book)

SIDE NOTE

$Z_1 + Z_2 = 6\ \Omega$ is purely real (+j4 and −j4 cancel), which is why both cases are clean. Equal and opposite reactances in series act like a short: **series resonance**, met later.

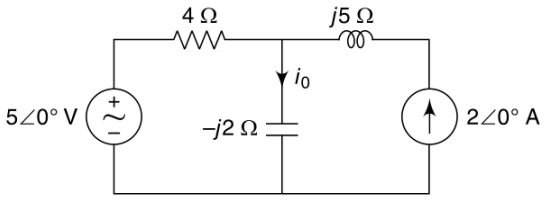
A5 · Problem 5.5

Superposition · AC, DC + AC **new idea · labels on new steps only**

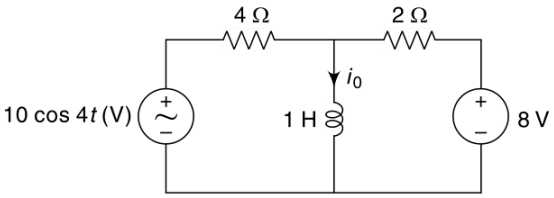
QUESTION

Find the current i_0 using the superposition theorem, in each of circuits (a), (b) and (c).

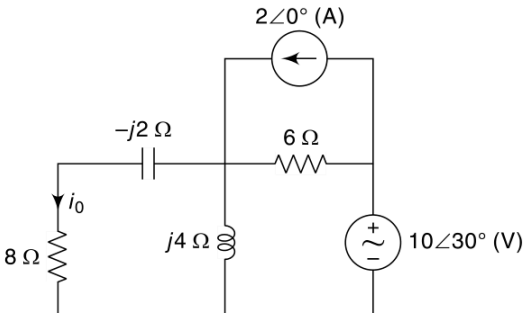
(a)



(b)



(c)



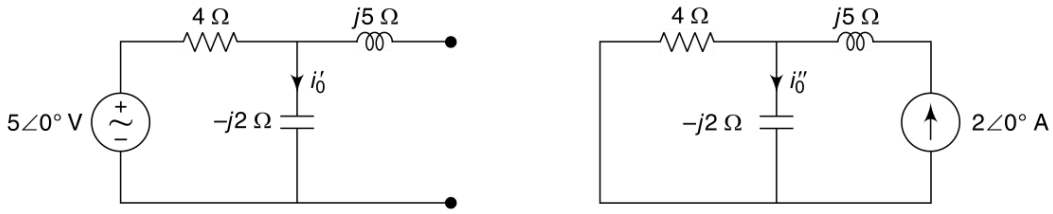
TRY IT FIRST

(a) is quick practice. In (b), stop and ask: what does an inductor look like to the 8 V **DC** source? In (c), redraw the circuit after switching off each source before you calculate anything.

HINT: WHAT'S NEW

- **(b) Two different frequencies (DC and 10 cos 4t).** Each case is solved **at its own frequency**: at DC ($\omega = 0$), $X_L = \omega L = 0$, so the **inductor is a short**; at $\omega = 4$, it is j4 Ω. The two answers are added **in the time domain**, never as phasors, because phasors only add when they share one frequency.
- **(c) Redraw after switching off.** Once the voltage source is shorted, the 6 Ω and the current source both run to ground and everything becomes parallel.

(a) Given: 5∠0° V, 2∠0° A, 4 Ω, −j2 Ω, j5 Ω. **To find:** i_0 (down the capacitor).



(i) voltage source alone · (ii) current source alone

$$5\angle 0° \text{ alone:} \qquad i_0' = \frac{5}{4 - j2} = 1 + j0.5 \text{ A}$$

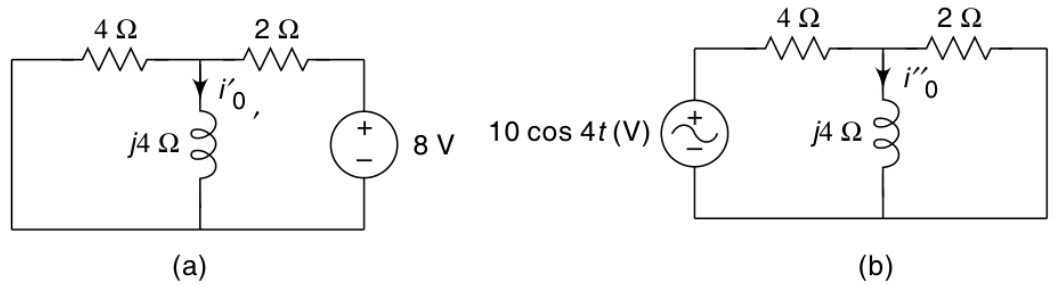
$2\angle 0°$ alone (j5 is in series with the source, so irrelevant):

$$i_0'' = 2 \times \frac{4}{4 - j2} = 1.6 + j0.8 \text{ A}$$

ANSWER

(a) $i_0 = 2.6 + j1.3 = 2.91\angle 26.57°$ A (= book, $2.9\angle 26.56°$ A)

(b) Given: 10 cos 4t V, 8 V DC, 4 Ω, 1 H, 2 Ω. **To find:** $i_0(t)$ (down the inductor).



The book's sub-circuits (a) DC alone, (b) AC alone. In (a) the inductor must be read as a short circuit.

HOW 8 V DC alone (AC source shorted): ω = 0, so $X_L = \omega L = 0$ and the inductor is a wire.

$$i_0' = \frac{8}{2} = 4 \text{ A} \qquad \text{(the 4 Ω has 0 V across it)}$$

WHAT Solve the DC case with the inductor replaced by a short.

WHY An impedance belongs to one frequency. Each case is solved at **its own source's** ω, and for DC, ω = 0. With the inductor a wire, the middle node is at 0 V; the 4 Ω's other end is the shorted AC source, also 0 V, so it carries nothing. The 8 V drives the 2 Ω alone.

10 cos 4t alone (8 V shorted; ω = 4 rad/s, $Z_L = j4$ Ω):

$$Z = 4 + \frac{j4 \times 2}{2 + j4} = 5.6 + j0.8\ \Omega, \qquad I = \frac{10\angle 0°}{Z} = 1.75 - j0.25 \text{ A}$$

$$i_0'' = I \times \frac{2}{2 + j4} = 0.25 - j0.75 = 0.791\angle -71.57^\circ \text{ A}$$

HOW

$$i_0(t) = 4 + 0.791 \cos(4t - 71.57^\circ) \text{ A}$$

WHAT Turn each part into a time function, then add.

WHY A phasor stands for a cosine at **one fixed ω**. The DC part doesn't rotate at all; the AC part rotates at 4 rad/s. Adding them as complex numbers would treat both as the same frequency, which they are not. In the time domain they add with no problem.

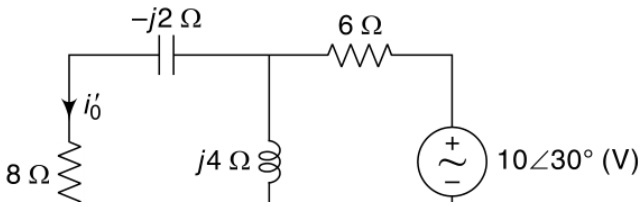
ANSWER

(b) $i_0(t) = 4 + 0.791 \cos(4t - 71.57^\circ)$ A

BOOK ERROR IN 5.5(B)

The book treats the inductor as j4 Ω for the **DC** source too, and then adds the DC and AC results as if both were phasors, printing $2.63\angle -63.58^\circ$ A. Both steps are wrong: at DC an inductor is a short, and quantities at different frequencies cannot be added as phasors. The answer above was confirmed by a node solve of each case.

(c) Given: $10\angle 30^\circ$ V, $2\angle 0^\circ$ A, 8 Ω, −j2 Ω, j4 Ω, 6 Ω. **To find:** i_0 (down the 8 Ω).



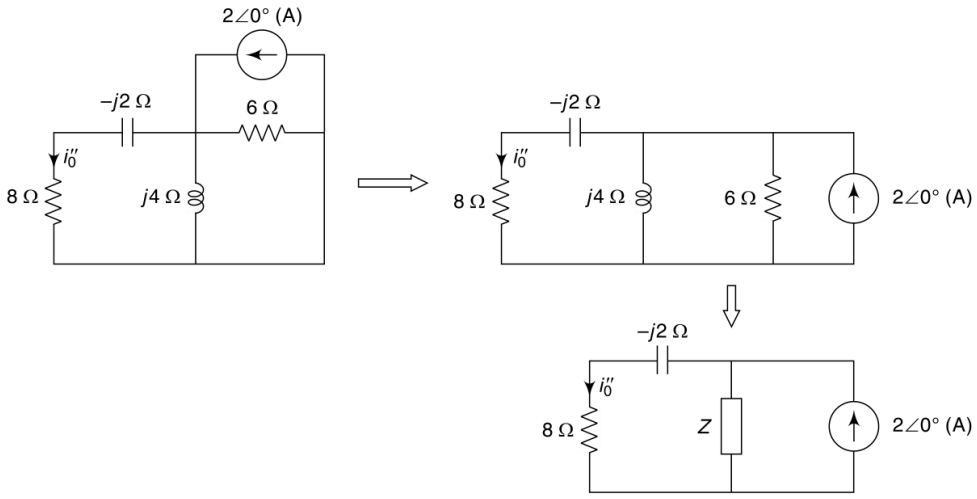
(i) voltage source alone

$10\angle 30^\circ$ V alone (current source opened):

$$Z = 6 + j4 \parallel (8 - j2) = 6 + (1.882 + j3.529) = 7.882 + j3.529 \ \Omega,$$

$$I = \frac{10\angle 30^\circ}{Z} = 1.158\angle 5.88^\circ \text{ A}$$

$$i_0' = I \times \frac{j4}{(8 - j2) + j4} = I \times \frac{j4}{8 + j2} = 0.562\angle 81.84^\circ \text{ A}$$



(ii) current source alone, redrawn: the 6 Ω and the source now both go to ground

HOW 2∠0° A alone (voltage source shorted). Redrawn: the source, the 6 Ω, the j4 and the (8 − j2) branch all join the same top node to ground. Z = j4 ‖ 6 = 1.846 + j2.769 Ω:

$$i_0'' = 2 \times \frac{Z}{(8 - j2) + Z} = \frac{j24}{28 + j22} = 0.674\angle 51.84^\circ \text{ A}$$

WHAT Redraw after shorting the voltage source, merge j4 ‖ 6 into Z, then a current divider between Z and the (8 − j2) branch.

WHY The shorted source turns the 6 Ω's right end into ground, which the original drawing hides. Once redrawn it is "a current source feeding parallel branches", a known current shared between paths, which is exactly a divider. Merging two branches first leaves a two-way split.

$$i_0 = (0.080 + j0.556) + (0.416 + j0.530) = 0.496 + j1.086 \text{ A}$$

ANSWER

(c) $i_0 = 1.19\angle 65.45^\circ$ A

BOOK ERROR IN 5.5(C)

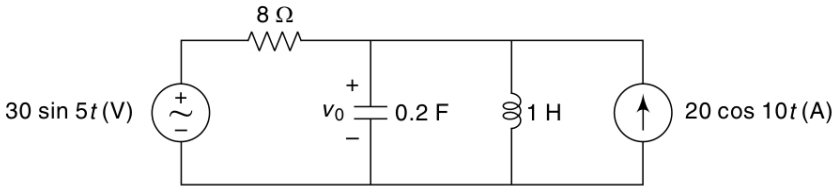
The book prints 0.81∠39.38° A. Two slips: in case (i) it divides the current with (8 − j2)/(8 + j2) instead of j4/(8 + j2) (the current into a branch is set by the **other** branch's impedance); in case (ii) it expands (8 − j2)(3 + j2) + j12 as 12 + j11 instead of 28 + j22. The answer above was confirmed by a full node solve.

A6 · Problem 5.6

Superposition · two frequencies **new idea · labels on new steps only**

QUESTION

Find v_0 using the superposition theorem.

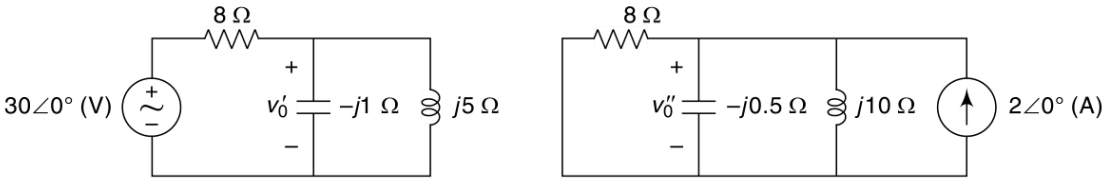


TRY IT FIRST

Work out X_C and X_L separately for ω = 5 and ω = 10 before drawing anything.

HINT: WHAT'S NEW

- **Reactances change with frequency.** $X_C = \frac{1}{\omega C}$ and $X_L = \omega L$ must be recomputed for each source's own ω.
- **Mixed sin and cos:** keep each term in the function it started with (sin for the voltage source, cos for the current source).



(i) voltage source alone, ω = 5 · (ii) current source alone, ω = 10

Given: 30 sin 5t V with 8 Ω; 0.2 F, 1 H, 20 cos 10t A. **To find:** $v_0(t)$.

HOW 30 sin 5t alone (ω = 5): $X_C = \frac{1}{5 \times 0.2} = 1\ \Omega \rightarrow -j1\ \Omega$; $X_L = 5 \times 1 = 5\ \Omega \rightarrow j5\ \Omega$. 20 cos 10t alone (ω = 10): $X_C = \frac{1}{10 \times 0.2} = 0.5\ \Omega \rightarrow -j0.5\ \Omega$; $X_L = 10\ \Omega \rightarrow j10\ \Omega$.

WHAT Compute each reactance twice, once per source frequency.

WHY The figure gives L and C, not impedances. An impedance only exists at a given ω, and here each case has a different ω, so each case gets its own impedances. Using one set for both cases is the common mistake.

30 sin 5t alone (current source opened), node equation at the top:

$$\frac{30 - v'}{8} = \frac{v'}{-j1} + \frac{v'}{j5} \quad \Rightarrow \quad v' = \frac{30}{8(0.125 + j0.8)} = 4.631\angle -81.12°\ \text{V}$$

20 cos 10t alone (voltage source shorted):

$$20 = v''\left(\frac{1}{8} + \frac{1}{-j0.5} + \frac{1}{j10}\right) = v''(0.125 + j1.9) \quad \Rightarrow \quad v'' = 10.50\angle -86.24°\ \text{V}$$

HOW

$$v_0(t) = 4.631 \sin(5t - 81.12°) + 10.50 \cos(10t - 86.24°) \text{ V}$$

WHAT Write each phasor back as a time function in its own original form (sin or cos), then add.

WHY A phasor only records size and angle; the sin or cos it came from is up to you to remember. The first source was a sine, so its answer is a sine; the second was a cosine. Different ω, so add in time, as in 5.5(b).

ANSWER

$v_0(t) = 4.631 \sin(5t - 81.12°) + 10.50 \cos(10t - 86.24°)$ V

BOOK INCONSISTENCY IN 5.6

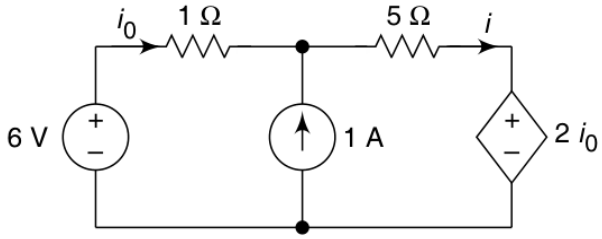
The figure gives the current source as **20** cos 10t A, but the book's solution uses **2**∠0° A and prints 1.051 cos(10t − 86.24°) for the second term. The answer above follows the figure. If your source is 2 A, the second term is 1.051 cos(10t − 86.24°) V; the first term is unchanged.

A7 · Problem 5.7

Superposition · current-controlled source **new idea · labels on new steps only**

QUESTION

Find i_0 and i from the circuit of the figure using the superposition theorem.



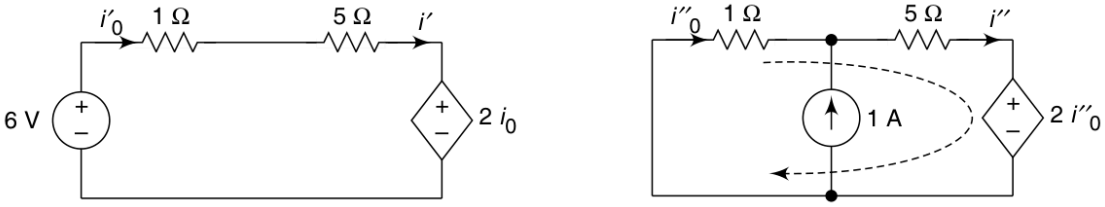
$2i_0$ is a current-controlled voltage source.

TRY IT FIRST

In each case, the dependent source's value is 2 × (that case's i_0). Write KCL at the middle node and one KVL around the outer loop.

HINT: WHAT'S NEW

- **Controlled by a current, not a voltage** (5.3 was voltage-controlled). The rule is the same: keep the source, and re-find its controlling current i_0 in each case.



(i) 6 V alone · (ii) 1 A alone

Given: 6 V, 1 A, 1 Ω, 5 Ω, CCVS $2i_0$. **To find:** i_0, i.

6 V alone (1 A opened): one loop, so $i_0' = i'$.

$$6 = 1\cdot i' + 5\cdot i' + 2i_0' = 8i' \quad \Rightarrow \quad i' = i_0' = 0.75 \text{ A}$$

1 A alone (6 V shorted):

HOW KCL at the middle node: $i'' = 1 + i_0''$ outer KVL: $1\cdot i_0'' + 5i'' + 2i_0'' = 0$

WHAT KCL at the middle node (where the 1 A enters), then KVL around the outer loop, with the diamond worth $2i_0''$.

WHY Same two moves as 5.3. KCL where the known source current enters ties the two unknown currents together. For the second equation, take the loop that **avoids the 1 A source**: the voltage across a current source is unknown, so a loop through it would add a third unknown. The diamond is valued with **this case's** controlling current, i_0''.

$$3i_0'' + 5(1 + i_0'') = 0 \quad \Rightarrow \quad 8i_0'' = -5 \quad \Rightarrow \quad i_0'' = -0.625 \text{ A}, \quad i'' = 0.375 \text{ A}$$

ANSWER

$i = 0.75 + 0.375 = 1.125$ A, $i_0 = 0.75 - 0.625 = 0.125$ A

BOOK ERROR IN 5.7

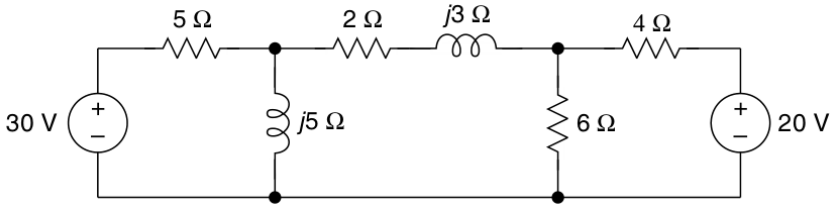
The book sets up the same equation, then solves $8i_0'' = -5$ as $i_0'' = -5/4 = -1.25$ A, and prints $i = 0.5$ A, $i_0 = -0.5$ A. The correct value is $-5/8 = -0.625$ A. Direct check: KCL $i = i_0 + 1$ and KVL $6 - i_0 - 5i = 2i_0$ give $i_0 = 1/8$ A ✓.

A8 · Problem 5.8

Superposition · AC **practice · How only**

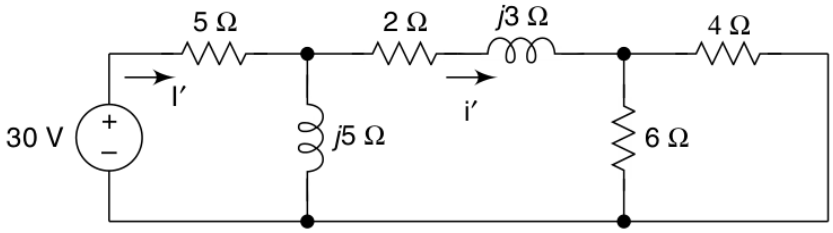
QUESTION

Using the superposition theorem, calculate the current through the (2 + j3) Ω impedance branch.



TRY IT FIRST

Series-parallel reduction, twice. Watch which way the branch current flows in case II.

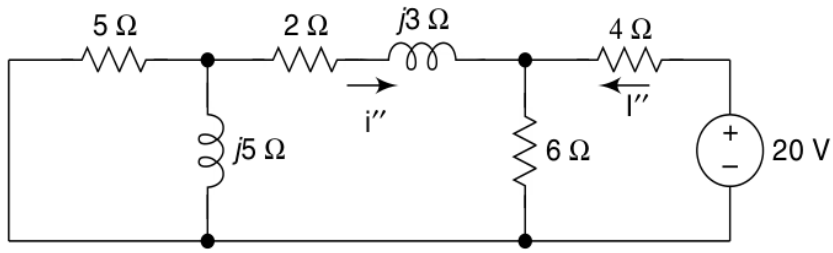


Case I: 30 V alone

Case I: 30 V alone (20 V shorted). Branch: $Z_b = (2 + j3) + (6 \parallel 4) = 4.4 + j3\ \Omega$.

$$Z = 5 + \frac{Z_b \times j5}{Z_b + j5} = 6.32 + j2.60\ \Omega, \qquad I' = \frac{30}{Z} = 4.06 - j1.67\ \text{A},$$

$$i' = I' \times \frac{j5}{Z_b + j5} = 2.389 + j0.270\ \text{A}$$



Case II: 20 V alone

Case II: 20 V alone (30 V shorted). Branch: $Z_a = (5 \parallel j5) + 2 + j3 = 4.5 + j5.5\ \Omega$.

$$Z = 4 + \frac{6Z_a}{Z_a + 6} = 7.31 + j1.41\ \Omega, \qquad I'' = \frac{20}{Z} = 2.638 - j0.509\ \text{A}$$

$$i'' = -I'' \times \frac{6}{Z_a + 6} = -1.063 + j0.848\ \text{A} \qquad \text{(it flows right → left, against i)}$$

ANSWER

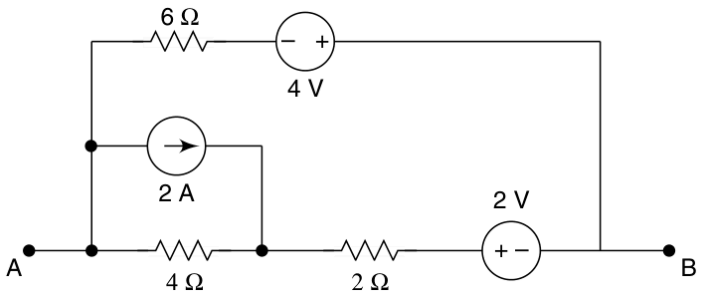
$i = i' + i'' = 1.326 + j1.117 = 1.73\angle 40.13°$ A (= book)

A9 · Problem 5.9

Superposition · three sources, open terminals **new idea · labels on new steps only**

QUESTION

Using the superposition theorem, find V_{AB}.

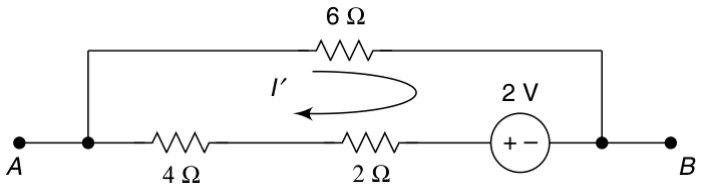


TRY IT FIRST

Three sources, so three cases. Nothing is connected across A–B, so no current leaves at A or B.

HINT: WHAT'S NEW

- **Three sources:** same method, one more case.
- **Open terminals:** V_{AB} is the voltage across a gap. With no load, the only path is the loop 4 Ω + 2 Ω + 6 Ω, and V_{AB} can be read along either side of that loop.
- **Source transformation inside a case:** 2 A in parallel with 4 Ω equals an 8 V source in series with 4 Ω, which makes case III a single loop.



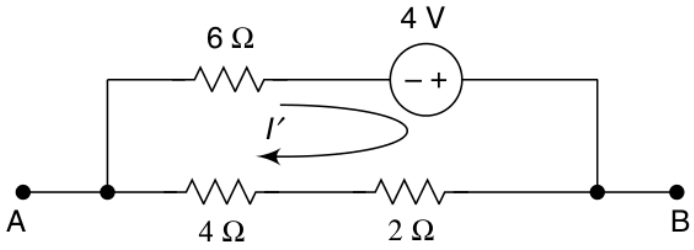
Case I: 2 V alone

HOW

Case I (2 V): $I' = \dfrac{2}{4+2+6} = \dfrac{1}{6}$ A, $V_{AB}' = 6I' = 1$ V

WHAT One loop current, then V_{AB} read across the top 6 Ω (its 4 V is switched off).

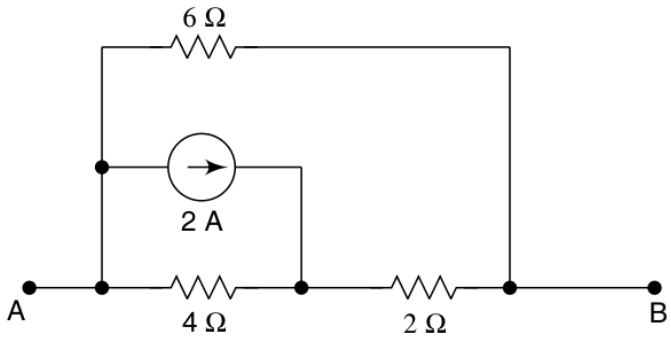
WHY Nothing is connected at A or B, so no current leaves there: the circuit is a single loop. V_{AB} is the voltage between two points, and it is the same along any path from A to B. Pick the path you know best: in this case the top path is just the 6 Ω. The current runs through it from A's side to B's side, so A is higher and the sign is +.



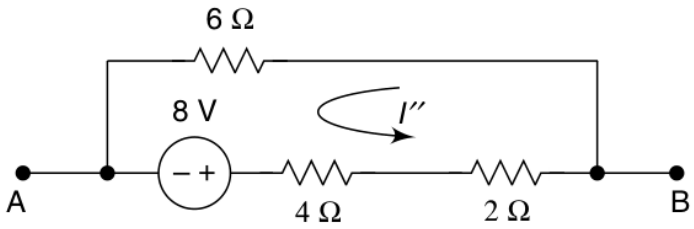
Case II: 4 V alone

Case II (4 V): $I'' = \frac{4}{12} = \frac{1}{3}$ A, $V_{AB}'' = -(4 + 2)I'' = -2$ V

(bottom path; current flows B → A along it)



Case III: 2 A alone



Case III after source transformation: 2 A ‖ 4 Ω → 8 V in series with 4 Ω

HOW 2 A ‖ 4 Ω → 8 V in series with 4 Ω, then one loop:

Case III (2 A): $I''' = \frac{8}{12} = \frac{2}{3}$ A, $V_{AB}''' = -6I''' = -4$ V

WHAT Source-transform the 2 A and its 4 Ω, then Ohm's law on the single loop.

WHY A current source in parallel with a resistor acts on the rest of the circuit exactly like V = IR in series with the same R. Converting it removes the only node where current splits, so the case becomes one loop (one equation, no KCL needed). Current now runs through the 6 Ω from B's side to A's side, hence the minus.

ANSWER

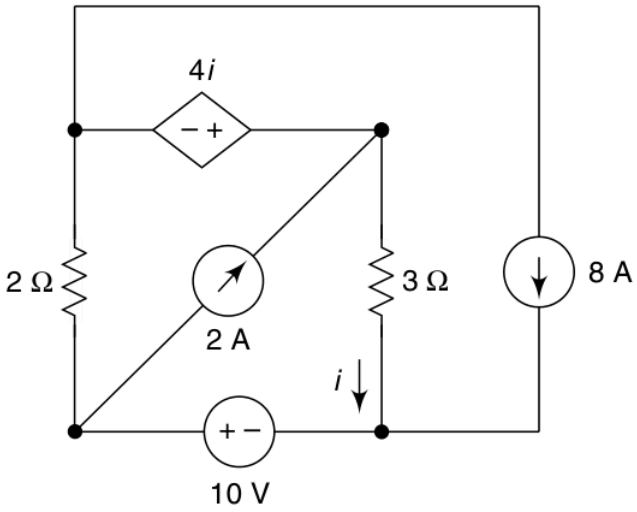
V_{AB} = 1 − 2 − 4 = −5 V (= book)

A10 · Problem 5.10

Superposition · dependent source, three sources **new idea · labels on new steps only**

QUESTION

Find the current i in the circuit shown using the superposition theorem.



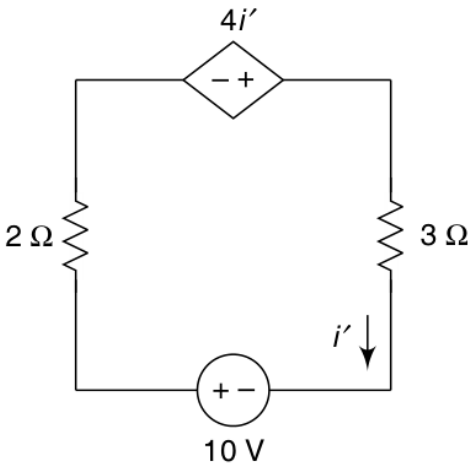
4i is a current-controlled voltage source; i flows down the 3 Ω.

TRY IT FIRST

Case I is one loop. In case II, look for a resistor in series with the dependent source.

HINT: WHAT'S NEW

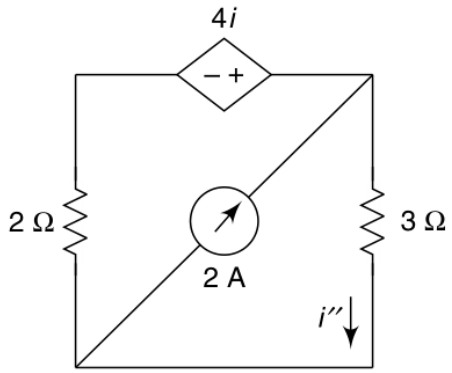
- **Source transformation works on dependent sources too.** In case II, the 4i source in series with the 2 Ω becomes a dependent current source 4i/2 = **2i** in parallel with 2 Ω.
- **A current source inside a mesh fixes that mesh current** (case III: the 8 A source sets the mesh difference I − 8).



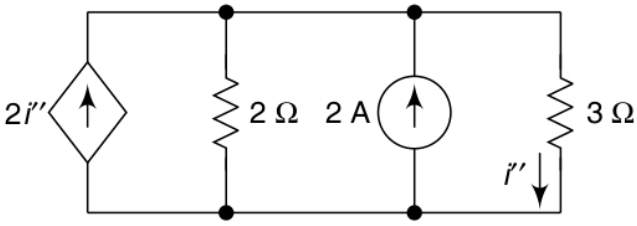
Case I: 10 V alone

Case I: 10 V alone (2 A and 8 A opened). Loop current i′ clockwise:

$$-4i' + 3i' - 10 + 2i' = 0 \quad \Rightarrow \quad i' = 10\ \text{A}$$



Case II: 2 A alone



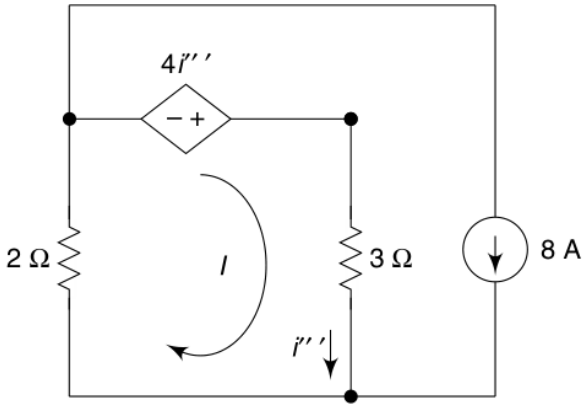
Case II after converting 4i″ + 2 Ω into 2i″ ‖ 2 Ω

HOW 4i″ in series with 2 Ω → 2i″ in parallel with 2 Ω. Now a total of (2 + 2i″) splits between 2 Ω and 3 Ω:

$$i'' = \frac{2}{2+3}(2 + 2i'') \quad \Rightarrow \quad 5i'' = 4 + 4i'' \quad \Rightarrow \quad i'' = 4\ \text{A}$$

WHAT Source-transform the dependent source with its series 2 Ω, then a current divider between 2 Ω and 3 Ω.

WHY With the 8 A opened, the left node has only two elements, the 2 Ω and the diamond, so they are in series and can be transformed (V ÷ R = 4i″ ÷ 2). Everything then hangs between the same two nodes, so the whole case is one divider. The value stays written in i″, the **controlling** current; it is solved for at the end.



Case III: 8 A alone

HOW Mesh current I clockwise in the inner loop; the 8 A is the outer mesh's current, so the shared 3 Ω carries $i''' = I - 8$:

$$-4i''' + 3(I - 8) + 2I = 0, \qquad I = i''' + 8 \qquad \Rightarrow \qquad -4i''' + 3i''' + 2i''' + 16 = 0$$

$$\Rightarrow \qquad i''' = -16 \text{ A}$$

WHAT Mesh analysis with one unknown mesh current; the 8 A fixes the other.

WHY A current source sitting alone on the outside edge of a mesh **is** that mesh's current: 8 A, known, so no equation is needed for it. In an element shared by two meshes, the current is the difference of the two mesh currents. That leaves one KVL for one unknown.

ANSWER

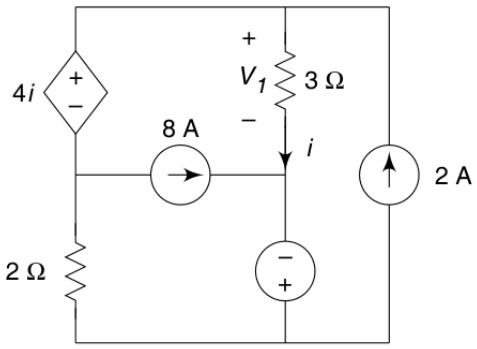
i = 10 + 4 − 16 = −2 A (= book)

A11 · Problem 5.11

Superposition · dependent source, three sources **practice · How only**

QUESTION

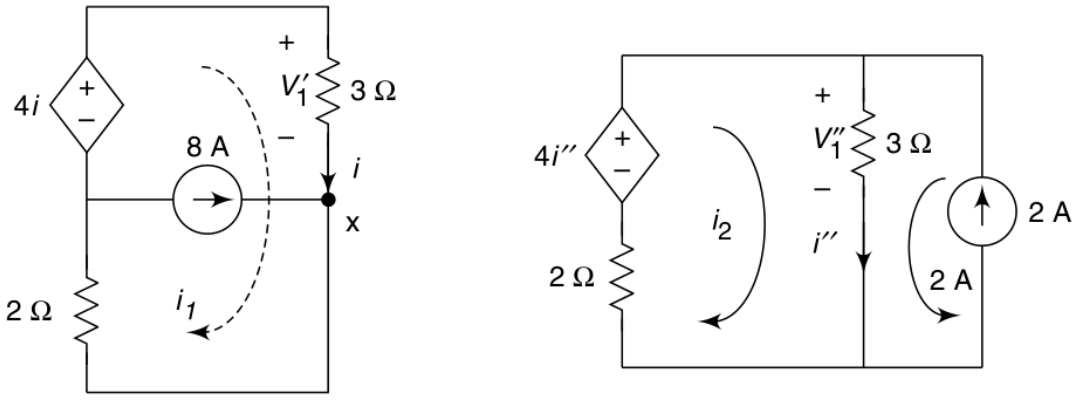
Using the superposition theorem, determine V_1, the voltage across the 3 Ω resistor.



The figure prints no value on the bottom-middle voltage source; the book's solution uses 10 V.

TRY IT FIRST

Three cases again; the dependent source 4i stays in all three. Case I has a supermesh around the 8 A source.



Case I: 8 A alone · Case II: 2 A alone

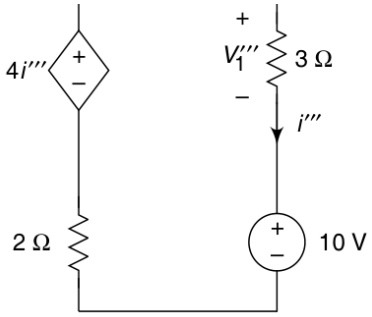
Case I: 8 A alone. Supermesh KVL and KCL at node x:

$$3i' + 2i_1 - 4i' = 0 \Rightarrow i_1 = \frac{i'}{2}; \qquad i_1 = 8 + i' \Rightarrow \frac{i'}{2} = 8 + i' \Rightarrow i' = -16\ \text{A},$$

$$V_1' = 3i' = -48\ \text{V}$$

Case II: 2 A alone. Mesh current i_2 on the left; the 3 Ω carries $i'' = i_2 + 2$:

$$3(i_2 + 2) + 2i_2 - 4i'' = 0 \Rightarrow 5i_2 + 6 - 4(i_2 + 2) = 0 \Rightarrow i_2 = 2\ \text{A}, \qquad i'' = 4\ \text{A}, \qquad V_1'' = 12\ \text{V}$$



Case III: 10 V alone

Case III: 10 V alone. One loop:

$$3i''' - 10 + 2i''' - 4i''' = 0 \Rightarrow i''' = 10\ \text{A}, \qquad V_1''' = 30\ \text{V}$$

ANSWER

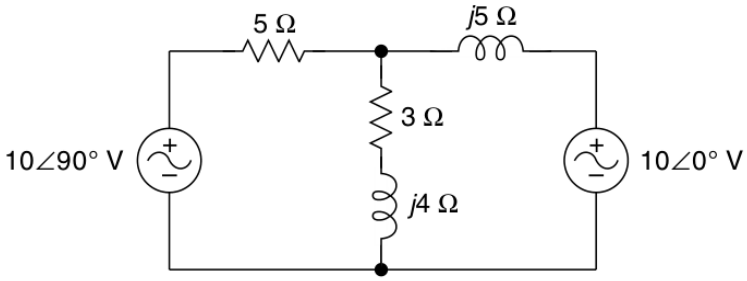
$V_1 = -48 + 12 + 30 = -6$ V (= book)

A12 · Problem 5.12

Superposition · AC **new idea · labels on new steps only**

QUESTION

Calculate the current through the impedance (3 + j4) Ω using the superposition theorem.

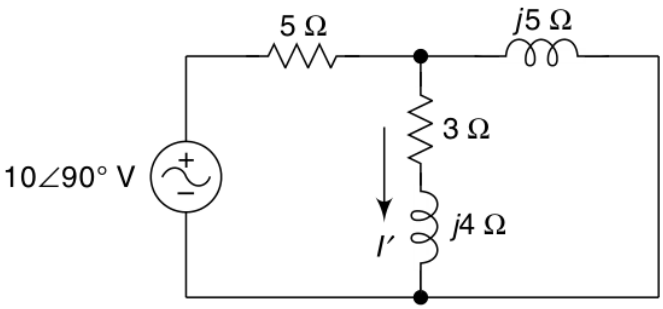


TRY IT FIRST

Two voltage sources, same frequency. Get each case's current as a fraction before simplifying.

HINT: WHAT'S NEW

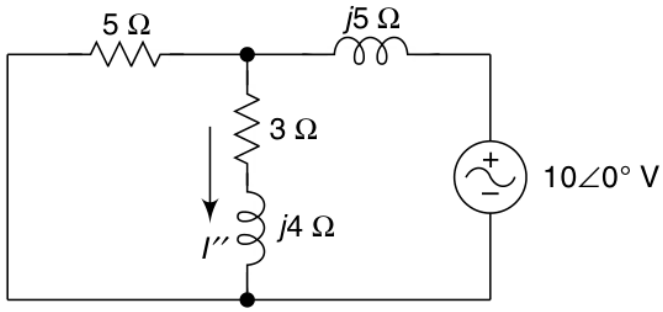
- **The two contributions can cancel.** Here they come out equal and opposite, so the branch current is zero. A zero answer is not an error; check it with one node equation.



10∠90° V alone

$$10\angle 90^\circ \text{ alone:} \quad I = \frac{j10}{5 + \dfrac{(3 + j4)j5}{3 + j9}} = \frac{j10(3 + j9)}{-5 + j60},$$

$$I' = I \times \frac{j5}{3 + j9} = \frac{-10}{-1 + j12} = 0.069 + j0.828 \text{ A}$$



10∠0° V alone

$$10\angle 0°\text{ alone:}\qquad I = \frac{10}{j5 + \dfrac{5(3+j4)}{8+j4}} = \frac{10(8+j4)}{-5+j60},$$

$$I'' = I \times \frac{5}{8+j4} = \frac{10}{-1+j12} = -0.069 - j0.828\ \text{A}$$

HOW

$$I = I' + I'' = 0$$

Check, node equation at the top node: $V(\frac{1}{5} + \frac{1}{3+j4} + \frac{1}{j5}) = \frac{j10}{5} + \frac{10}{j5} = j2 - j2 = 0$, so $V = 0$ ✓.

WHAT Add, then confirm the zero with one node equation.

WHY A zero can come from a real cancellation or from a sign slip, and the sum alone can't tell which. One independent check (nodal, all sources on) settles it in a line. Here each source pushes exactly j2 A into the node, from opposite sides.

ANSWER

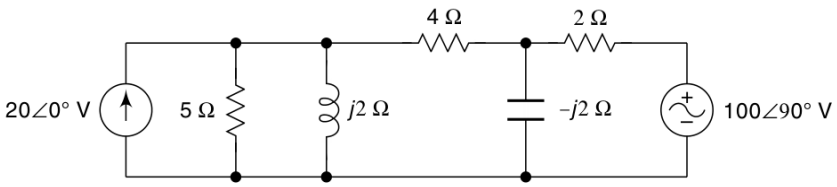
I = 0 A (= book)

A13 · Problem 5.13

Superposition · AC **practice · How only**

QUESTION

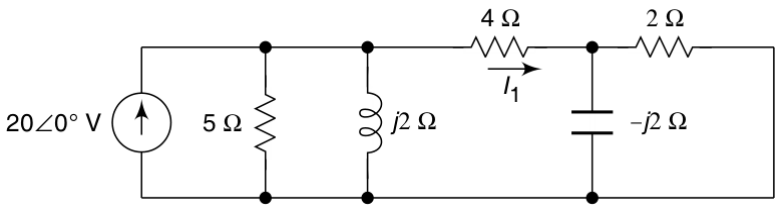
Using the superposition theorem, determine the current in the 4 Ω resistor.



The book labels the current source "20∠0° V"; it is a current source (arrow), 20∠0° A.

TRY IT FIRST

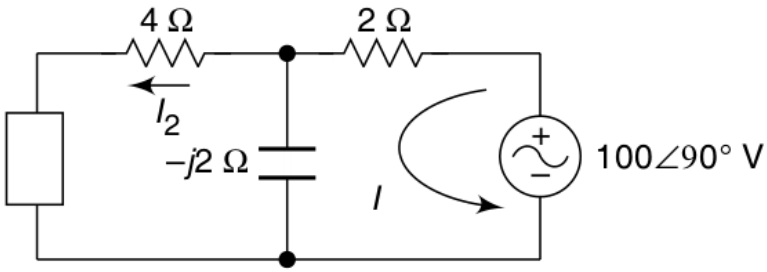
Combine the parallel pairs first: 5 Ω ∥ j2 Ω and 2 Ω ∥ −j2 Ω.



Case I: 20∠0° A alone

Case I: 20∠0° A alone (voltage source shorted). $Z_1 = 5 \parallel j2 = 0.690 + j1.724\ \Omega$; $Z_2 = 2 \parallel (-j2) = 1 - j1\ \Omega$.

$$I_1 = 20 \times \frac{Z_1}{Z_1 + 4 + Z_2} = 3.145 + j5.660 \text{ A} \qquad \text{(left → right)}$$



Case II: 100∠90° V alone

Case II: 100∠90° V alone (current source opened). $(Z_1 + 4) \parallel (-j2) = 0.850 - j1.950\ \Omega$:

$$I_s = \frac{j100}{2 + (0.850 - j1.950)} = -16.35 + j23.90 \text{ A},$$
$$V_B = I_s(0.850 - j1.950) = 32.70 + j52.20 \text{ V}$$

$$I_2 = \frac{V_B}{Z_1 + 4} = 9.748 + j7.547 \text{ A} \qquad \text{(right → left)}$$

ANSWER

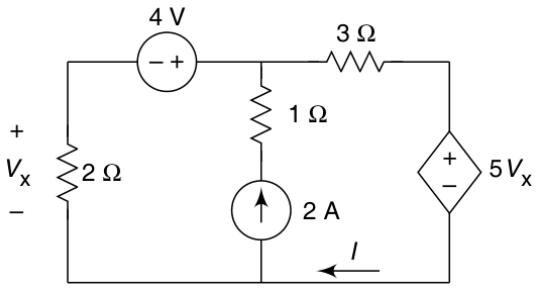
$I = I_1 - I_2 = -6.604 - j1.887 = 6.87\angle -164.05°$ A (left → right), i.e. 6.87 A flowing right → left (= book, 6.89∠−163.67° A, to rounding)

A14 · Problem 5.14

Superposition · dependent source **practice · How only**

QUESTION

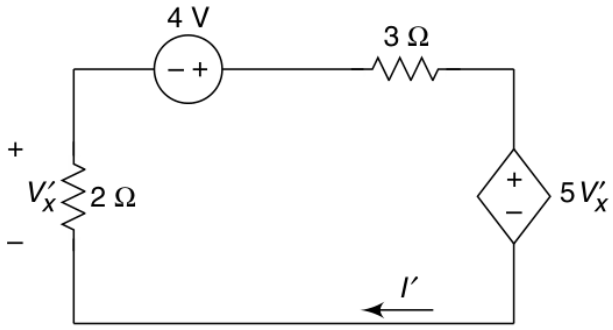
Find I in the figure using the superposition theorem.



$5V_x$ is a voltage-controlled voltage source; V_x is across the 2 Ω (+ at top).

TRY IT FIRST

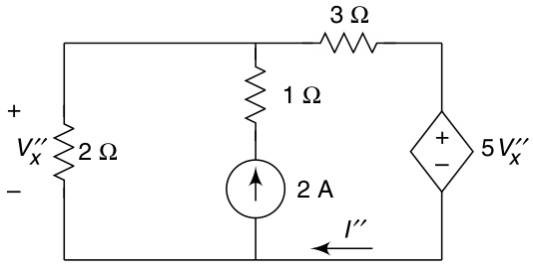
Same idea as 5.3. In case II, the 4 V short joins the two top-left nodes; write one KCL there.



Case I: 4 V alone

Case I: 4 V alone (2 A opened). One loop; I′ flows up the 2 Ω, so $V_x' = -2I'$:

$$-4 + 3I' + 5V_x' - V_x' = 0 \Rightarrow 3I' + 4(-2I') = 4 \Rightarrow I' = -0.8 \text{ A}$$



Case II: 2 A alone

Case II: 2 A alone (4 V shorted). KCL at the top node (voltage V_x''):

$$2 = \frac{V_x''}{2} + \frac{V_x'' - 5V_x''}{3} \Rightarrow V_x'' = -2.4 \text{ V}, \qquad I'' = \frac{V_x'' - 5V_x''}{3} = 3.2 \text{ A}$$

ANSWER

I = −0.8 + 3.2 = 2.4 A (= book)